VITISH 2026
(SIH Internal Hackathon)
TITLE PAGE

- **Problem Statement ID: PS17**
- **Problem Statement Title: Space Debris & Collision Avoidance**
- **Theme: Space Technology**
- **PS Category: Hardware**

IDEA TITLE

- **Edge-Vision Collision Avoidance & AI-Driven Error Ellipsoid Reduction**

- Ground AI: Predicts drag via NOAA solar flux to shrink CDM error.
- Edge Hardware: FPGA taps Star Trackers for real-time visual confirmation via frame-differencing.

- Eliminates false positives from radar "error ellipsoids".
- Preserves critical Δv propellant, extending satellite lifespan.

- Zero Added Mass: Repurposes existing optics.
- Hardware Isolation: Prevents vision logic from hijacking ADCS.
- Active-Active Protocol: Matrix negotiates right-of-way to prevent double-burns.

TECHNICAL APPROACH

- Hardware: FPGA (Altera DE-10), Star Tracker Optics
- Software: Python (XGBoost, Skyfield), Verilog, C/C++
- Data & UI: NOAA API, Space-Track.org, CesiumJS

SOLUTION FEASIBILITY & VIABILITY

- Highly feasible to rapidly prototype using a Hardware-in-the-Loop (HITL) testbench.
- Utilizes proven Commercial Off-The-Shelf (COTS) Star Trackers and standard FPGA (Altera DE-10) architecture.

- Optical Blindness: Debris cannot be tracked while in the Earth's shadow (eclipse).
- Streak Smearing: Head-on relative velocities (>15 km/s) spread point sources into unreadable faint streaks.

- Blindness Mitigation: Orbit propagator detects eclipse conditions and automatically defaults to conservative ground-radar estimates.
- Smearing Mitigation: FPGA runs parallelized Hough transforms tuned specifically for multi-frame transient line detection.

REAL WORLD IMPACTS, BENEFITS & RELEVANCE

IMPACT: Replaces guesswork with deterministic, onboard tracking data for commercial constellation operators and defense agencies.

BENEFITS:

- **Economic**: Saves critical **Delta V (ΔV)** propellant, directly delaying End-of-Life (EOL) decommissioning and extending the satellite's revenue-generating lifespan.
- **Operational**: Minimizes service downtime during conjunctions via the active-active maneuver negotiation protocol.
- **Environmental**: Actively prevents the runaway spread of Kessler Syndrome by tracking and avoiding otherwise untrackable micro-debris.

TEAM CAPABILITY

- Team's experience in the domain :-

Previously worked with FPGA,HDL,Flight control softwares,Orbital mechanics,PCBs and basic circuiting.

- Why choosing the domain :-

Relentless focus on aerospace dynamics and solving high-speed edge computing bottlenecks for orbital autonomy and propellant preservation.

RESEARCH AND REFERENCES

- Links of the litrature review: click here